

STATEMENT OF WORK
Project Number VBDZ262103
Replace Underground Water Valves

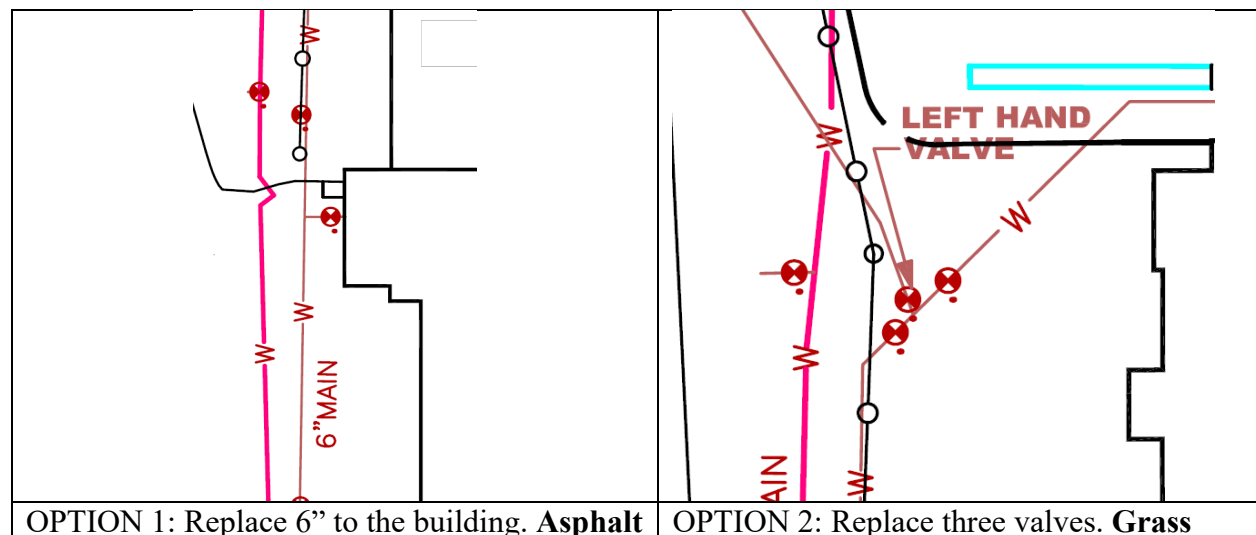
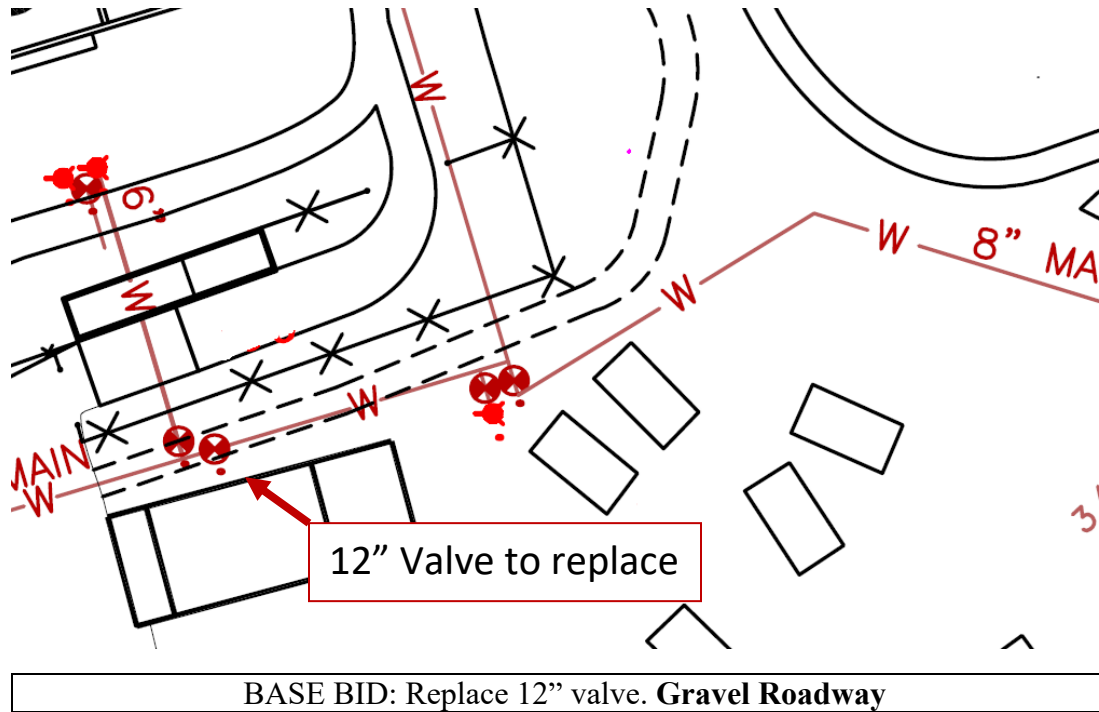
DESCRIPTION:

BASE BID: Water valve on at 12" pipe cannot fully shut down. The valve must be replaced. The utility map will be provided upon the contract award.

OPTION 1: replace 6" valve to the building. The valve is in the asphalt.

OPTION 2: replace three valves. The valves are in the grass.

OPTION 3: an elbow on an 8" line is leaking and needs replacement. The elbow is in the dirt.

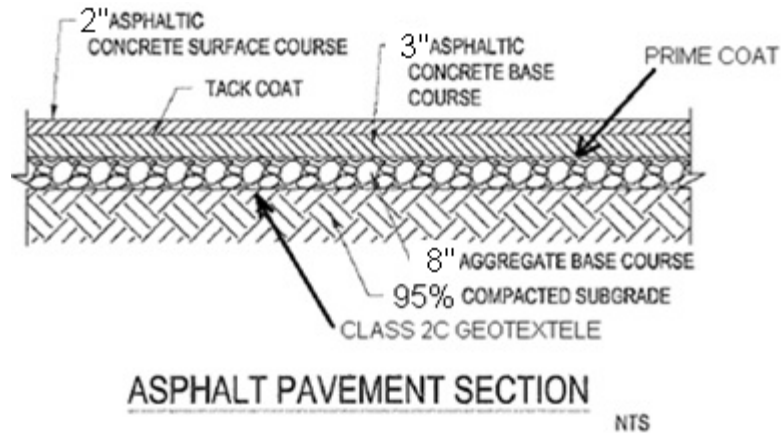


1. BASE SCOPE

- 1.1. Fill out the dig permit a week before starting the work. The paperwork includes calling Dig Safe and Base's internal procedures.
- 1.2. Dig out the earth around the valve. Expect to dig about 6 feet or deeper.
 - 1.2.1. If the hole is 5 feet or deeper, protective systems such as sloping, benching, shoring, or shielding shall be used.
- 1.3. To minimize operational impact, water utility shutdowns shall only be scheduled on designated installation 'down days' (which occur every other Friday). Once the water supply is shut off, the Contractor shall work continuously until the new valve is installed and fully functional. If the work is not completed by the end of the scheduled Friday, the Contractor is required to continue work uninterrupted through the weekend. Full water service must be restored and fully operational no later than the start of normal business hours on the following Monday. Note: Work performed under this specific requirement supersedes the standard Work Hours defined elsewhere in this SOW.
- 1.4. Coordinate with the base any water line portions isolation or complete water shut down.
- 1.5. Remove the old valve and dispose of it in a base metal dumpster.
- 1.6. Install a new 12" gate valve.
 - 1.6.1. Provide and install all the couplings, fittings, clamps, sleeves, gaskets and any other materials necessary to properly install the new unit.
 - 1.6.2. Provide valves and fittings with pressure ratings equivalent to the pressure ratings of the pipe.
- 1.7. Before placing in position, clean pipe, fittings, valves, and accessories, and maintain them in a clean condition. Provide proper facilities for lowering sections of pipe into trenches. Do not drop or dump pipe, fittings, valves, or other water line material into trenches. Cut pipe cleanly, squarely, and accurately to the length established at the site and work in place without springing or forcing.
- 1.8. Provide anchors and supports for fastening work into place. Make provision for the expansion and contraction of pipelines. Keep the trench free of water until joints have been assembled. At the end of each workday, close open ends of pipe temporarily with wood blocks or bulkheads. Do not lay pipe when conditions of trench or weather prevent installation.
- 1.9. Where wet or otherwise unstable soil incapable of properly supporting the valve, as determined by the Contracting Officer, is unexpectedly encountered at the bottom of a trench, such material shall be removed to the depth required and replaced to the proper grade with select granular material.
- 1.10. The bedding surface for the valve and the replaced portion of the pipe shall provide a firm foundation of uniform density throughout the entire length of the replaced portion of the pipe.
- 1.11. Backfill should be sand within 12" of the pipe and Class II for the remainder of the trench backfill in roadways or satisfactory materials in lawn areas (free of organic or frozen material and rocks with any dimension greater than 3").
- 1.12. After the valve and joint pipe have been properly bedded, selected material from excavation or borrow, at a moisture content that will facilitate compaction, shall be placed

along both sides of the pipe in layers not exceeding 6 inches in compacted depth. The backfill shall be brought up evenly on both sides of the pipe for the full length of the pipe. The fill shall be thoroughly compacted under the haunches of the pipe. Each layer shall be thoroughly compacted with mechanical tampers or rammers. This method of filling and compacting shall continue until the fill has reached an elevation equal to the midpoint (spring line) of the concrete pipe or has reached an elevation of at least 12 inches above the top of the pipe for flexible pipe. The remainder of the trench shall be backfilled and compacted by spreading and rolling or compacted by mechanical rammers or tampers in layers not exceeding 8 inches. Density tests shall be made as necessary to ensure conformance to the compaction requirements specified below. Where it is necessary, in the opinion of the Contracting Officer, that sheeting or portions of bracing be left in place, the contract will be adjusted accordingly. Untreated sheeting shall not be left in place beneath structures or pavements.

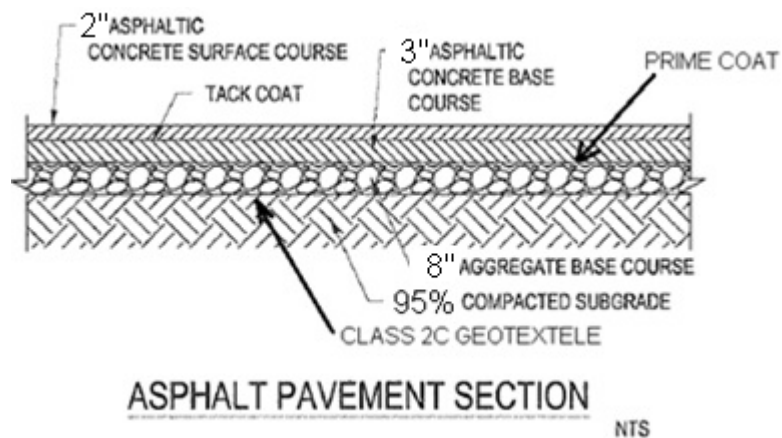
- 1.13. Compaction density shall be not less than 90 percent of maximum density for cohesive material and 95 percent of maximum density for cohesionless material.
- 1.14. Gate Valves on Buried Piping - AWWA C509 or AWWA C515. Unless otherwise specified, valves conforming to: (1) AWWA C509 or AWWA C515 shall be non-rising stem type with mechanical-joint ends. Valves shall open by counterclockwise rotation of the valve stem. Stuffing boxes shall have O-ring stem seals. Stuffing boxes shall be bolted and constructed so as to permit easy removal of parts for repair.
- 1.15. Valve Boxes - Provide a valve box for each gate valve on buried piping. Valve boxes shall be of cast iron of a size suitable for the valve on which it is to be used and shall be adjustable with extensions as required. Cast-iron boxes shall have a minimum cover and wall thickness of 3/16 inch. Provide a round head. Cast the word "WATER" on the lid. The least diameter of the shaft of the box shall be 5 1/4 inches. Cast-iron box shall have a heavy coat of bituminous paint. There shall be no leaks after the installation.
- 1.16. Inspect pipe and fittings prior to backfilling and repair as required. Contractor shall NOT backfill the valve until COR verifies that the valve does not leak.
- 1.17. Backfill and pave following the Asphalt Pavement Section figure below to meet the existing pavement.
- 1.18. Tack coat shall be applied between pavement courses as well as pavement lifts.
- 1.19.



- 1.20. Contractor is responsible for site cleanliness and removal of post-construction debris and site restoration.
- 1.21. Install and maintain perimeter sediment controls (e.g., silt fences, fiber rolls, or rock filters) down-slope of all disturbed areas prior to soil-disturbing activities.
- 1.22. Stabilize disturbed soils via seeding, mulching, erosion control blankets or any other means.
- 1.23. Protect storm drainage from any debris or sediment.

2. OPTION 1

- 2.1. Replace 6" valve leading to the building with a gate valve.
- 2.2. The same requirements as in the BASE BID apply.
- 2.3. Backfill and pave following the Asphalt Pavement Section figure below to meet the existing pavement.
- 2.4. Tack coat shall be applied between pavement courses as well as pavement lifts.

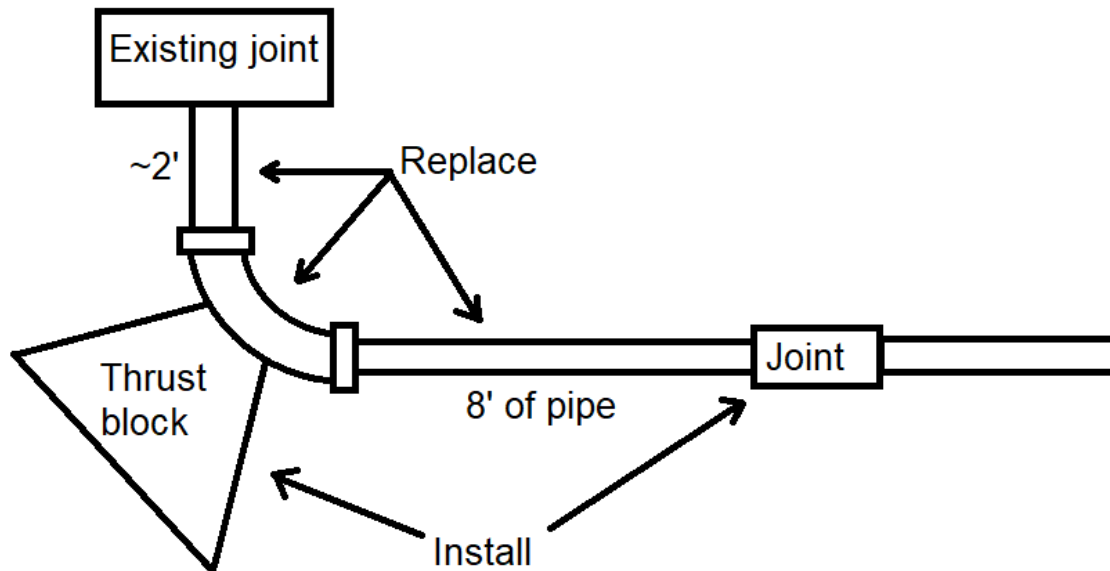


3. OPTION 2

- 3.1. Replace three (3) 6" valves in the grass with **right-handed** gate valves.
- 3.2. The same requirements as in the BASE BID apply.

4. OPTION 3

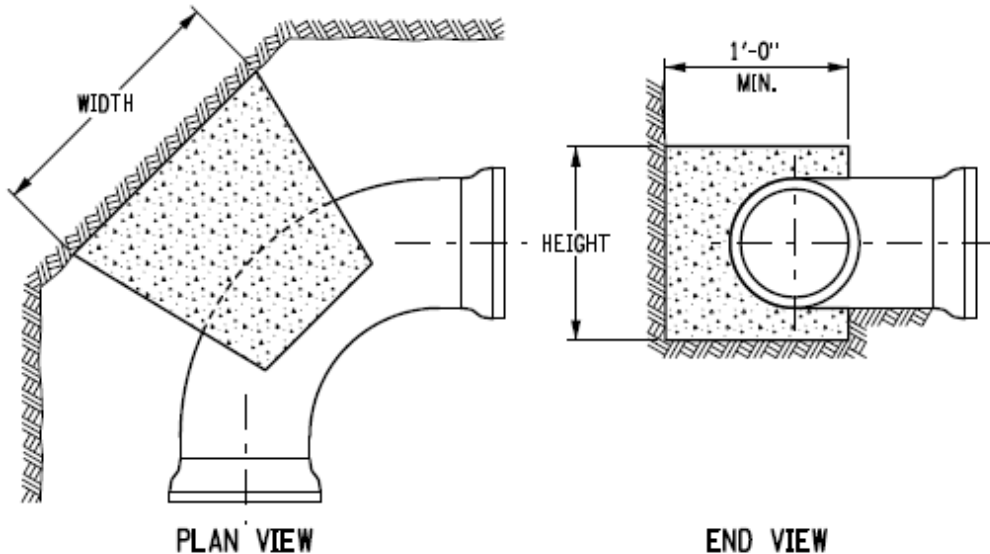
- 4.1. Contractor shall replace an elbow leading to the fire hydrant. The elbow is located ~100 ft from the valve in the base scope. The utilities map will be provided.



- 4.2. Contractor shall dig out the earth around the hydrant. Expect to dig about 6 feet or deeper.
- 4.3. Remove the elbow and cut out 2 feet of the pipe on one side of the elbow and 8 feet on the other side of the elbow.
- 4.4. Dispose of the metal in the base's metal dumpster.
- 4.5. Contractor shall remove the boulder touching the elbow.
- 4.6. Contractor shall install a new elbow, a new joint, and two new pipe segments as shown in the figure above.
- 4.6.1. The material to use is ductile iron 60-42-10 grade ceramic epoxy lined, Class 56. Mechanical and Push-on joints shall conform to ANSI Specification A21.11.
- 4.7. Each pipe shall be laid true to line and grade to form a close concentric joint with the adjoining pipe, preventing offsets in the flow line.
- 4.8. Contractor shall install a pipe footing to avoid sagging.
- 4.9. Contractor shall install a concrete thrust blocking. See figure below.
- 4.9.1. Precast concrete thrust block is allowed.
- 4.9.2. If poured at the spot, Use ASTM C94/C94M concrete having a minimum compressive strength of 3,000 psi at 28 days. Do not backfill or begin testing until at least 7 days after placing of the concrete.
- 4.10. Contractor shall test the line.
- 4.11. Contractor shall keep the excavation free from water during construction. No structure shall be built or pipe shall be laid in water, and water shall not be allowed to flow over or rise upon any concrete or pipe until the same has been inspected and the concrete or joint material has thoroughly set.
- 4.12. Backfill will not be allowed until the flow and pressure are restored and confirmed that the joints are not leaking. Backfilling shall be done in two (2) distinct operations. All

backfill beneath, around, and to a depth of twelve (12") inches above the top of the pipe shall be placed by hand in four (4") inch layers for the full width of the trench and thoroughly compacted by hand with vibratory equipment. The remainder of the backfill shall be placed in 6" layers and compacted to the top of the trench, either by pneumatic hand tamps, hydro-tamps, or other approved methods. Care shall be taken so that the pipe is not laterally displaced during backfilling.

- 4.13. Contractor shall place a pipe-location tape about 2' below grade.



90° BEND THRUST BLOCK DIMENSIONING					
PIPE SIZE	WIDTH (FT-IN)	HEIGHT (FT-IN)	PIPE SIZE	WIDTH (FT-IN)	HEIGHT (FT-IN)
4 NPS	2'-3"	1'-3"	14 NPS	7'-3"	3'-6"
6 NPS	3'-3"	1'-9"	16 NPS	8'-3"	4'-0"
8 NPS	4'-3"	2'-3"	18 NPS	9'-3"	4'-6"
10 NPS	5'-3"	2'-6"	20 NPS	10'-6"	5'-0"
12 NPS	6'-0"	3'-3"	24 NPS	12'-6"	6'-0"

5. PROPOSED PROJECT SCHEDULE

The Project shall begin 14 days after the Notice to Proceed is issued. The start date may be changed with approval of Contracting Officer. The project shall be completed within 60 days following the Notice to Proceed or agreed start date.

6. CONTRACTOR REQUIREMENTS

- 6.1. The contractor shall comply with all applicable installation/facility access and local security policies and procedures, which may be obtained from the Contracting Officer's Representative (COR). The contractor shall also provide all information required for background checks to meet installation access requirements to be accomplished by the local

installation's Security Forces, Director of Emergency Services or local Security Office. The contractor shall ensure compliance with all personal identity verification requirements as directed by DoD, HAF and/or local policy. Should the Force Protection Condition (FPCON) change, the Government may require changes in contractor security matters or processes.

- 6.2. Contractor and all associated sub-contractors employees shall comply with adjudication standards and procedures using the National Crime Information Center Interstate Identification Index (NCIC-III) and Terrorist Screening Database (TSDB) (DODIO 2000.16V1_DAFI 31-145, DAFI 31-101 and DODM 5200.08v3_DAFMAN 31-101v3), applicable installation, facility and area commander installation/facility access and local security policies and procedures (provided by government representative), or, at OCONUS locations, in accordance with status of forces agreements and other theater regulations.
- 6.3. All contractor employees, to include subcontractor employees, requiring access to US Government installations, facilities and controlled access areas shall complete AT Level I training within 30 calendar days after contract start date or effective date of incorporation of this requirement into the contract, whichever is applicable. AT Level I may be accomplished by a Level I qualified instructor; completing the Force Protection computer-based training (CBT) course on Joint Knowledge Online (JKO) at <https://jkodirect.jten.mil/Atlas2/page/login/Login.jsf>. For those without a CAC, select "Non-CAC" user on the JKO site. Tracking for each contractor or subcontractor employee is the responsibility of the COR or unit AT Representative.
- 6.4. Requirement for OPSEC Training. In accordance with AFI 10-701, Operations Security, OPSEC PMs/SMO/SMNCOs/Coordinators will provide OPSEC training or training materials to contract employees within 30 days of employees' initial assignment to the contract and prior to having access to any critical information (CI) and annually thereafter. General organizational orientations may need to be supplemented by duty-related orientations in the work center targeted toward specific critical information and vulnerabilities associated with the work. Initial training (OPSE 1301 - OPSEC Fundamentals) located on the Defense Counterintelligence and Security Agency website: <https://securityawareness.usalearning.gov/opsec/index.htm> and the OPSEC Annual training- DAF-Operations Security Awareness Training located on My Learning <https://lms-jets.cce.af.mil/moodle/course/view.php?id=11654#section-1>.
- 6.5. The Contractor shall provide all personnel and necessary equipment to establish an adequate and safe work zone. OSHA is applicable at the site and inspectors appear unexpectedly.
- 6.6. Contractor is responsible for site cleanliness and shall remove any waste or debris after the work is complete. All materials shall be secured to prevent loose objects from blowing and hindering flying operations in any manner.
- 6.7. The Contractor shall recycle to the greatest extent possible and turn in weigh slips to 109 Environmental Management Office for construction debris.
- 6.8. The Contractor shall bring any unforeseen project conditions to the attention of the COR immediately. Unexpected or differing site conditions, which may alter the scope of work, shall be approved by the contracting officer (KO) prior to such work being performed.
- 6.9. The Contractor must get approval from the COR prior to any utility shutdowns. (do not contact production control directly)

- 6.10. The Contractor shall notify the COR prior to delivery of any material or equipment to the site. The contractor shall ensure someone is available to receive delivered materials and able to store them safely in an area to be coordinated with the government project manager. Any materials delivered without prior coordination and without having someone to take receipt, shall not be accepted and turned away.
- 6.11. Contractor shall repair or replace in kind any property that is damaged.
- 6.12. Contractor hours are 0700-1530 Monday thru Friday, excluding Government holidays. No work shall be performed on holidays.